University of St.Gallen
Institute for Accounting, Controlling and Auditing

Module 7: Capital Budgeting and Cost Analysis

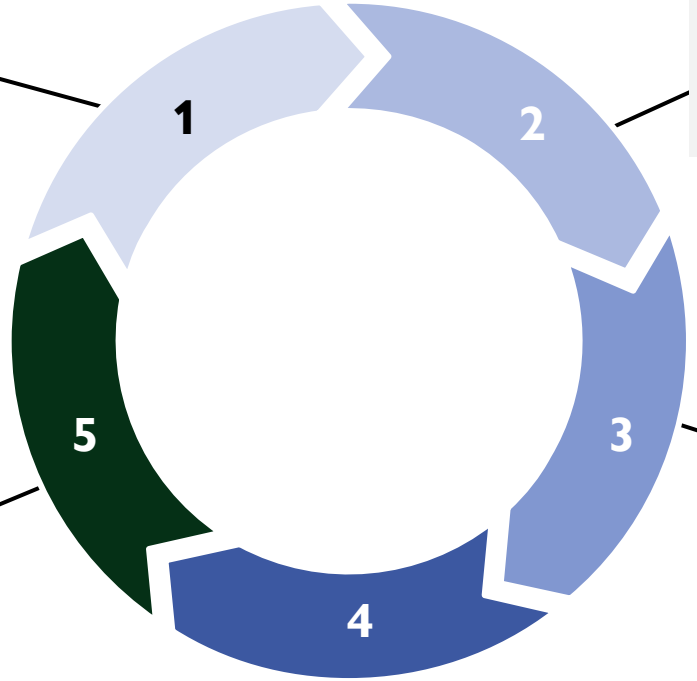
Module 8: Balanced Scorecard

Module 9: Performance Management

Accounting, Controlling, Auditing (ACA)
Week 3 – Tutorial

From insight to impact.

Five Stages of Capital Budgeting



1. Indentify projects – identify potential capital investments that agree with the organization's strategy

2. Obtain information- gather information from all parts of the value chain to evaluate alternative projects

3. Make predictions- Forecast all potential cash flows attributable to the alternative projects

4. Make Decisions by Choosing Among Alternatives – Determine which investment yields the greatest benefit and the least cost to the organization

5. Implement the decision, evaluate performance, and learn (generally separated into two phases):
Obtain funding and make the investment selected in stage 4
Track realized cashflows, compare against estimated numbers, and revise plans if necessary

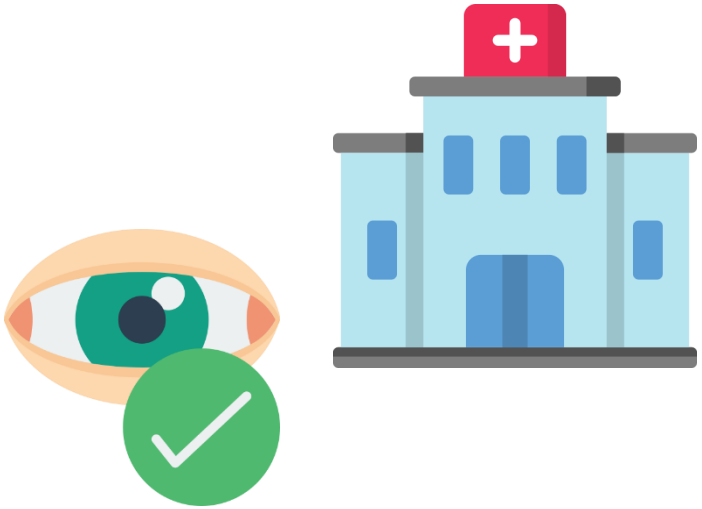
CTRL 2020, Ch. 11, p. 413 ff.

Methods of Capital Budgeting

Net Present Value Method (NPV)	**Internal Rate of Return Method** (IRR)	**Payback Method**	**Accrual Accounting Rate of Return Method** (AARR)
▪ The NPV method calculates the expected profit or loss of a project by discounting all expected future cash flows (DCF) to the present (t_0) ▪ The discount rate used is the Required Rate of Return (RRR). The RRR corresponds to the minimum return with which an investor wants to be compensated for assuming the risks of a project. ↳ *Only projects with a positive NPV or NPV = 0 are acceptable*	▪ The internal rate of return method corresponds to the discount rate at which the present value of all cash flows from a specific project is zero. In other words, the IRR is the discount rate at which the NPV = 0. ▪ The internal rate of return is based on the same formula as the NPV. ↳ *A project is only acceptable if the IRR is at least equal to the RRR*	▪ The payback method calculates the time needed for a project or investment to recover the (start-up) losses ▪ The payback method is aimed at liquidity ↳ *Managers usually prefer projects with a short payback period*	▪ The AARR method divides the average annual return of a project by the initial investment ↳ *The AARR method is similar to the IRR method, as both methods calculate a percentage value as return*

Task 1: Capital Budgeting Methods

City Hospital, a nonprofit organization, estimates that it can save **CHF 28,000 a year in cash operating costs** for the next **10 years** if it buys a special-purpose eye-testing machine at a **cost of CHF 110,000**. **No terminal disposal value is expected.** City Hospital's **required rate of return is 14%.** Assume all cash flows occur **at year-end** except for **initial investment amounts**. City Hospital uses **straight-line depreciation.**



Task 1: Capital Budgeting Methods

Calculate the following for the special-purpose eye-testing machine:

- Net present value
- Payback period
- Internal rate of return
- Accrual accounting rate of return based on net initial investment
- Accrual accounting rate of return based on average investment

Methods of Capital Budgeting

Net Present Value Method (NPV)	**Internal Rate of Return Method** (IRR)	**Payback Method**	**Accrual Accounting Rate of Return Method** (AARR)
▪ The NPV method calculates the expected profit or loss of a project by discounting all expected future cash flows (DCF) to the present (t_0) ▪ The discount rate used is the Required Rate of Return (RRR). The RRR corresponds to the minimum return with which an investor wants to be compensated for assuming the risks of a project. ↳ *Only projects with a positive NPV or NPV = 0 are acceptable*	▪ The internal rate of return method corresponds to the discount rate at which the present value of all cash flows from a specific project is zero. In other words, the IRR is the discount rate at which the NPV = 0. ▪ The internal rate of return is based on the same formula as the NPV. ↳ *A project is only acceptable if the IRR is at least equal to the RRR*	▪ The payback method calculates the time needed for a project or investment to recover the (start-up) losses ▪ The payback method is aimed at liquidity ↳ *Managers usually prefer projects with a short payback period*	▪ The AARR method divides the average annual return of a project by the initial investment ↳ *The AARR method is similar to the IRR method, as both methods calculate a percentage value as return*

Task 1: Capital Budgeting

Initial investment	**110’000**
Useful life	**10 years**
Annual cashflow	**28’000**
RRR	**14%**

Net Present Value

Time value of cashflows	Time value of 1 CHF	Useful Life (Cashflow in TCHF)										
		0	1	2	3	4	5	6	7	8	9	10
-110’000.00 CHF	1.000	110										
24’561.40 CHF	0.8772		28									
21’545.09 CHF	0.7695			28								
18’899.20 CHF	0.6750				28							
16’578.25 CHF	0.5921					28						
14’542.32 CHF	0.5194						28					
12’756.42 CHF	0.4556							28				
11’189.85 CHF	0.3996								28			
9’815.65 CHF	0.3506									28		
8’610.22 CHF	0.3075										28	
7’552.83 CHF	0.2697											28
36’051.24 CHF												

$$\mathbf{NPV = -I_0 + \frac{CF_n}{(1+r)^n}}$$

Task 1: Capital Budgeting

Initial investment	**110’000**
Useful life	**10 years**
Annual cashflow	**28’000**
RRR	**14%**

Net Present Value

Alternative calculation

Present value of an annuity for n payment periods: Uniform, regular cash flow that flows over a fixed number of periods.

Present Value of Ordinary Annuity of $1

$$PV_A = \frac{1}{i}\left[1 - \frac{1}{(1+i)^n}\right]$$

Periods	3%	4%	5%	6%	7%	8%	10%	12%	14%
1	.9709	.9615	.9524	.9434	.9346	.9259	.9091	.8929	.8772
2	1.9135	1.8861	1.8594	1.8334	1.8080	1.7833	1.7355	1.6901	1.6467
3	2.8286	2.7751	2.7232	2.6730	2.6243	2.5771	2.4869	2.4018	2.3216
4	3.7171	3.6299	3.5460	3.4651	3.3872	3.3121	3.1699	3.0373	2.9137
5	4.5797	4.4518	4.3295	4.2124	4.1002	3.9927	3.7908	3.6048	3.4331
6	5.4172	5.2421	5.0757	4.9173	4.7665	4.6229	4.3553	4.1114	3.8887
7	6.2303	6.0021	5.7864	5.5824	5.3893	5.2064	4.8684	4.5638	4.2883
8	7.0197	6.7327	6.4632	6.2098	5.9713	5.7466	5.3349	4.9676	4.6389
9	7.7861	7.4353	7.1078	6.8017	6.5152	6.2469	5.7590	5.3282	4.9464
10	8.5302	8.1109	7.7217	7.3601	7.0236	6.7101	6.1446	5.6502	5.2161
11	9.2526	8.7605	8.3064	7.8869	7.4987	7.1390	6.4951	5.9377	5.4527
12	9.9540	9.3851	8.8633	8.3838	7.9427	7.5361	6.8137	6.1944	5.6603
13	10.6350	9.9856	9.3936	8.8527	8.3577	7.9038	7.1034	6.4235	5.8424
14	11.2961	10.5631	9.8986	9.2950	8.7455	8.2442	7.3677	6.6282	6.0021
15	11.9379	11.1184	10.3797	9.7122	9.1079	8.5595	7.6061	6.8109	6.1422

Cashflow 28’000 CHF x **Annuity 5.2161** - **Investment 110’000 CHF**

= **Net Present Value 36’048 CHF**

Methods of Capital Budgeting

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Task 1: Capital Budgeting

Initial investment	**110’000**
Useful life	**10 years**
Annual cashflow	**28’000**
RRR	**14%**

Payback Method

Equal cashflows

$$\text{Payback period} = \frac{\text{Initial Investment}}{\text{Cashflow p.a.}}$$

$$\frac{\text{Initial Investment 110’000 CHF}}{\text{Cashflow p.a. 28’000 CHF}} = 3.93 \text{ years}$$

Methods of Capital Budgeting

Net Present Value Method (NPV)	**Internal Rate of Return Method** (IRR)	**Payback Method**	**Accrual Accounting Rate of Return Method** (AARR)
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Task 1: Capital Budgeting

Accrual Accounting Rate of Return Method (AARR)

Initial investment	**110’000**
Useful life	**10 years**
Annual cashflow	**28’000**
RRR	**14%**

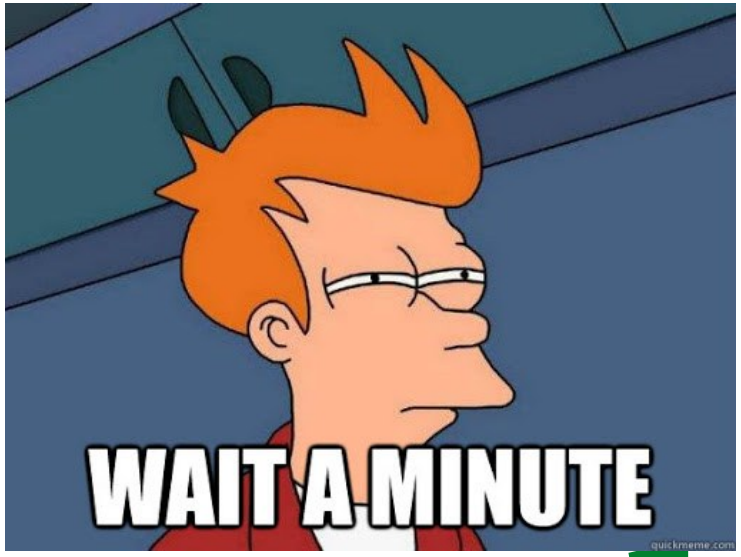
Based on net initial investment

Constant Net Income

$$AARR = \frac{\text{Expected net income}}{\text{Initial investment}}$$

Annual straight-line depreciation → $\frac{\text{Initial investment } 110'000 \text{ CHF}}{\text{Useful life } 10 \text{ years}}$ = **Annual straight-line depreciation 11’000 CHF**

AARR → $\frac{\text{Constant Net Cashflow } 28'000 \text{ CHF} - \text{Annual Depreciation } 11'000 \text{ CHF}}{\text{Initial investment } 110'000 \text{ CHF}}$ = **15.45 %**



Task 1: Capital Budgeting

Accrual Accounting Rate of Return Method (AARR)

Initial investment	**110'000**
Useful life	**10 years**
Annual cashflow	**28'000**
RRR	**14%**

Based on average investment

Constant Net Income

$$\text{AARR} = \frac{\text{Expected net income}}{\text{Initial investment}}$$

Book value of the investment declines over time

Annual straight-line depreciation →

$$\frac{\text{Initial investment } 110'000 \text{ CHF}}{\text{Useful life } 10 \text{ years}} = \text{Annual straight-line depreciation } 11'000 \text{ CHF}$$

Average investment over 10 years →

$$\frac{\text{Net initial investment } 110'000 \text{ CHF} + \text{Net terminal cash flow } 0 \text{ CHF}}{2} = \text{Average investment } 55'000$$

AARR →

$$\frac{\text{Constant Net Income } 28'000 \text{ CHF} - \text{Annual Depreciation } 11'000 \text{ CHF}}{\text{Average investment } 55'000 \text{ CHF}} = 30.91\ \%$$

Task 1: Capital Budgeting

Factors City Hospital should **consider include:**

Quantitative Financial Aspects.

- Financing factors, such as the availability of cash to purchase the new equipment.

Qualitative Factors:

- Qualitative factors, such as the benefits to its customers of a better eye-testing machine
- The employee-morale advantages of having up-to-date equipment.
- …

Task 2: Capital Budgeting

Swiss Cola is considering the purchase of a **special-purpose bottling machine for CHF 65,000**. It is expected to have a useful life of **4 years with no terminal disposal value**. The plant manager estimates the following savings in cash operating costs:

Year	Amount
1	CHF 25,000
2	CHF 22,000
3	CHF 21,000
4	CHF 20,000
Total	CHF 88,000

Swiss Cola uses a required rate of return of **18% in its capital budgeting decisions**. Ignore income taxes in your analysis. Assume all cash flows occur at year-end except for initial investment amounts.

Task 2: Capital Budgeting

Calculate the following for the special-purpose bottling machine:

- Net present value
- Payback period
- Discounted payback period
- Accrual accounting rate of return based on net initial investment. (*Assume straight-line depreciation. Use the average annual savings in cash operating costs when computing the numerator of the accrual accounting rate of return.*)

Task 2: Capital Budgeting

Initial investment	**65'000**
Useful life	**4 years**
Annual cashflow	**Uneven**
RRR	**18%**

Net Present Value

Time value of cashflows	Time value of 1 CHF	Useful Life (Cashflow in TCHF)				
		0	1	2	3	4
-65'000.00 CHF	1.000	- 65'000				
21'175 CHF	0.847		25'000			
15'796 CHF	0.718			22'000		
12'789 CHF	0.609				21'000	
10'320 CHF	0.516					20'000
- 4'920 CHF						

Net Present Value

$$NPV = -I_0 + \frac{CF_n}{(1+r)^n}$$

Task 2: Capital Budgeting

Payback Method

Initial investment	**65'000**
Useful life	**4 years**
Annual cashflow	**Uneven**
RRR	**18%**

Uneven cashflows

Year	Cash Savings	Cumulative Cash Savings	Initial Investment Yet to Be Recorded at End of Year
0	-	-	CHF 65'000
1	CHF 25'000	CHF 25'000	CHF 40'000
2	CHF 22'000	CHF 47'000	CHF 18'000
3	CHF 21'000	CHF 68'000	-

$$\text{Payback period} = 2 \text{ years} + \frac{\text{CHF } 18'000}{\text{CHF } 21'000} = 2.86 \text{ years}$$

Task 2: Capital Budgeting

Discounted Payback Method

Initial investment	**65'000**
Useful life	**4 years**
Annual cashflow	**Uneven**
RRR	**18%**

Uneven cashflows

Year	Cash Savings	Discount Factor (18%)	Discounted Cash Savings	Cumulative Discounted Cash Savings	Unrecovered Investment
0	-	-	-	-	CHF 65'000
1	CHF 25'000	0.847	CHF 21'175	CHF 21'175	CHF 43'825
2	CHF 22'000	0.718	CHF 15'796	CHF 36'971	CHF 28'029
3	CHF 21'000	0.609	CHF 12'789	CHF 49'760	CHF 15'240
4	CHF 20'000	0.516	CHF 10'320	CHF 60'080	- CHF 4'920

At a 18% rate of return, this project does not save enough to make it worthwhile using the discounted payback method.

Task 2: Capital Budgeting

Accrual Accounting Rate of Return Method (AARR)

Initial investment	**65'000**
Useful life	**4 years**
Annual cashflow	**Uneven**
RRR	**18%**

Based on net initial investment

Average annual savings in cash operating costs

Average annual savings in cash operating costs → $\frac{\text{CHF 88'000}}{\text{4 years}}$ = **CHF 22'000**

Annual straight-line depreciation → $\frac{\text{CHF 65'000}}{\text{4 years}}$ = **CHF 16'250**

Accrual accounting rate of return → $\frac{\text{CHF 22'000} - \text{CHF 16'250}}{\text{CHF 65'000}} = \frac{\text{CHF 5'750}}{\text{CHF 65'000}} = 8.85\%$

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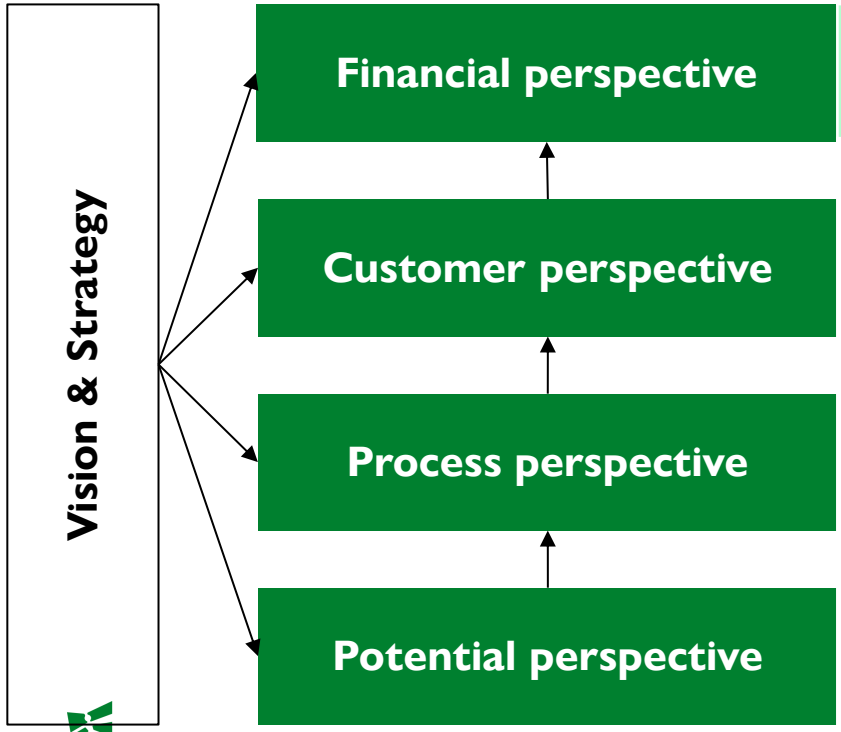
Balanced Scorecard (BSC) as a Solution?

Idea and structure:

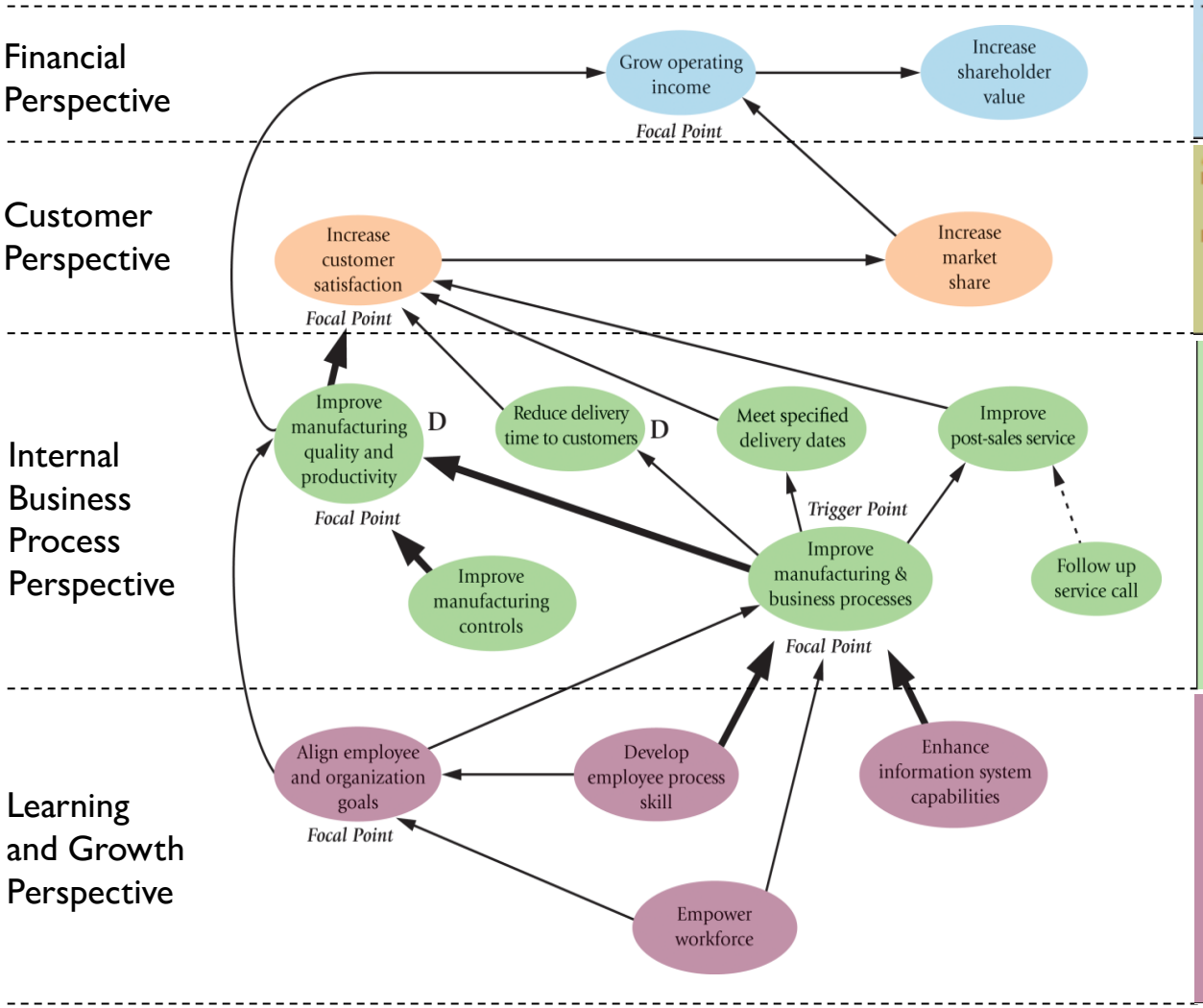
- BSC as an instrument for strategic implementation
- Leading and lagging measures
- Financial and non-financial measures
- Qualitative and quantitative measures



Strategic Goals	Measures	Target values	Activities
▪ Guidelines for influencing behavior ▪ Convert the strategy into action-oriented statements ▪ Example: Build a competitive cost structure	▪ Clear and unmistakable expression of the strategic goals ▪ Measurement of goal achievement ▪ Example: Total costs in% of the planned sales	▪ Target values should be demanding, ambitious and credibly achievable ▪ Example: 65%	▪ Measures, projects, programs and initiatives to implement the strategic goals ▪ Example: Optimize supply chain

Kapian, Norton; Baianced Scorecard (1996).

Strategy Map und Balanced Scorecard

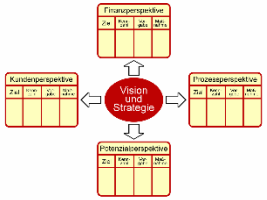


Strategic Objectives	Measures	Initiatives	Target Performance	Actual Performance
Financial Perspective				
Grow operating income	Operating income from productivity gain	Manage costs and unused capacity	$1,850,000	$1,912,500
Increase shareholder value	Operating income from growth	Build strong customer relationships	$2,500,000	$2,920,000
	Revenue growth		9%	10%[a]
Customer Perspective				
Increase market share	Market share in communication-networks segment	Identify future needs of customers	6%	7%
Increase customer satisfaction	Number of new customers	Identify new target-customer segments	1	1[b]
	Customer-satisfaction ratings	Increase customer focus of sales organization	90% of customers give top two ratings	87% of customers give top two ratings
Internal-Business-Process Perspective				
Improve postsales service	Service response time	Improve customer-service process	Within 4 hours	Within 3 hours
Improve manufacturing quality and productivity	Yield	Identify root causes of problems and improve quality	78%	79.3%
Reduce delivery time to customers	Order-delivery time	Reengineer order-delivery process	30 days	30 days
Meet specified delivery dates	On-time delivery	Reengineer order-delivery process	92%	90%
Improve processes	Number of major improvements in manufacturing and business processes	Organize teams from manufacturing and sales to modify processes	5	5
Improve manufacturing capability	Percentage of processes with advanced controls	Organize R&D/manufacturing teams to implement advanced controls	75%	75%
Learning-and-Growth Perspective				
Align employee and organization goals	Employee-satisfaction ratings	Employee participation and suggestions program to build teamwork	80% of employees give top two ratings	88% of employees give top two ratings
Empower workforce	Percentage of line workers empowered to manage processes	Have supervisors act as coaches rather than decision makers	85%	90%
Develop process skill	Percentage of employees trained in process and quality management	Employee training programs	90%	92%
Enhance information-system capabilities	Percentage of manufacturing processes with real-time feedback	Improve online and offline data gathering	80%	80%

[a](Revenues in 2013 – Revenues in 2012) ÷ Revenues in 2012 = ($25,300,000 – $23,000,000) ÷ $23,000,000 = 10%.
[b]Number of customers increased from seven to eight in 2013.

CTRL 2020, Ch. 9, p. 332.

Task 3: Balanced Scorecard



Following is a random-order listing of **perspectives, strategic objectives, and performance measures for the balanced scorecard**.

Perspectives
Internal business process
Customer
Learning and growth
Financial

Strategic Objectives
Acquire new customers
Increase shareholder value
Retain customers
Improve manufacturing quality
Develop profitable customers
Increase proprietary products
Increase information-system capabilities
Enhance employee skills
On-time delivery by suppliers
Increase profit generated by each salesperson
Introduce new products
Minimize invoice-error rate

Performance Measures
Percentage of defective-product units
Return on assets
Number of patents
Employee turnover rate
Net income
Customer profitability
Percentage of processes with real-time feedback
Return on sales
Average job-related training-hours per employee
Return on equity
Percentage of on-time deliveries by suppliers
Product cost per unit
Profit per salesperson
Percentage of error-free invoices
Customer cost per unit
Earnings per share
Number of new customers
Percentage of customers retained

Task 3: Balanced Scorecard

1. For each perspective, select those **strategic objectives from the list that best relate to it**. For each **strategic objective**, select the **most appropriate performance measure(s)** from the list.

Task 3: Balanced Scorecard

Perspectives	
Internal business process	P
Customer	C
Learning and growth	L
Financial	F

Strategic Objectives	
Acquire new customers	C
Increase shareholder value	F
Retain customers	C
Improve manufacturing quality	P
Develop profitable customers	C
Increase proprietary products	P
Increase information-system capabilities	L
Enhance employee skills	L
On-time delivery by suppliers	P
Increase profit generated by each salesperson	F
Introduce new products	P
Minimize invoice-error rate	P

Task 3: Balanced Scorecard

Strategic Objectives
Acquire new customers
Increase shareholder value
Retain customers
Improve manufacturing quality
Develop profitable customers
Increase proprietary products
Increase information-system capabilities
Enhance employee skills
On-time delivery by suppliers
Increase profit generated by each salesperson
Introduce new products
Minimize invoice-error rate

Performance Measures
Percentage of defective-product units
Return on assets
Number of patents
Employee turnover rate
Net income
Customer profitability
Percentage of processes with real-time feedback
Return on sales
Average job-related training-hours per employee
Return on equity
Percentage of on-time deliveries by suppliers
Product cost per unit
Profit per salesperson
Percentage of error-free invoices
Customer cost per unit
Earnings per share
Number of new customers
Percentage of customers retained

Task 3: Balanced Scorecard – Full Solution

Perspectives	Strategic Objectives	Performance Measures
▪ Financial	▪ Increase shareholder value	▪ Earnings per share ▪ Net income ▪ Return on assets ▪ Return on sales ▪ Return on equity ▪ Product cost per unit
	▪ Increase profit generated by each salesperson	▪ Profit per salesperson
▪ Customer	▪ Acquire new customers ▪ Retain customers ▪ Develop profitable customers	▪ Number of new customers ▪ Percentage of customers retained ▪ Customer profitability ▪ Customer cost per unit
▪ Internal Business Process	▪ Improve manufacturing quality ▪ Introduction of new products	▪ Percentage of defective product units ▪ Number of new products
	▪ Minimize invoice error rate ▪ On-time delivery by suppliers	▪ Percentage of error-free invoices ▪ Percentage of on-time deliveries by suppliers
	▪ Increase proprietary products	▪ Number of patents
▪ Learning and Growth	▪ Increase information system capabilities ▪ Enhance employee skills	▪ Percentage of processes with real-time feedback ▪ Employee turnover rate ▪ Average job-related training hours per employee

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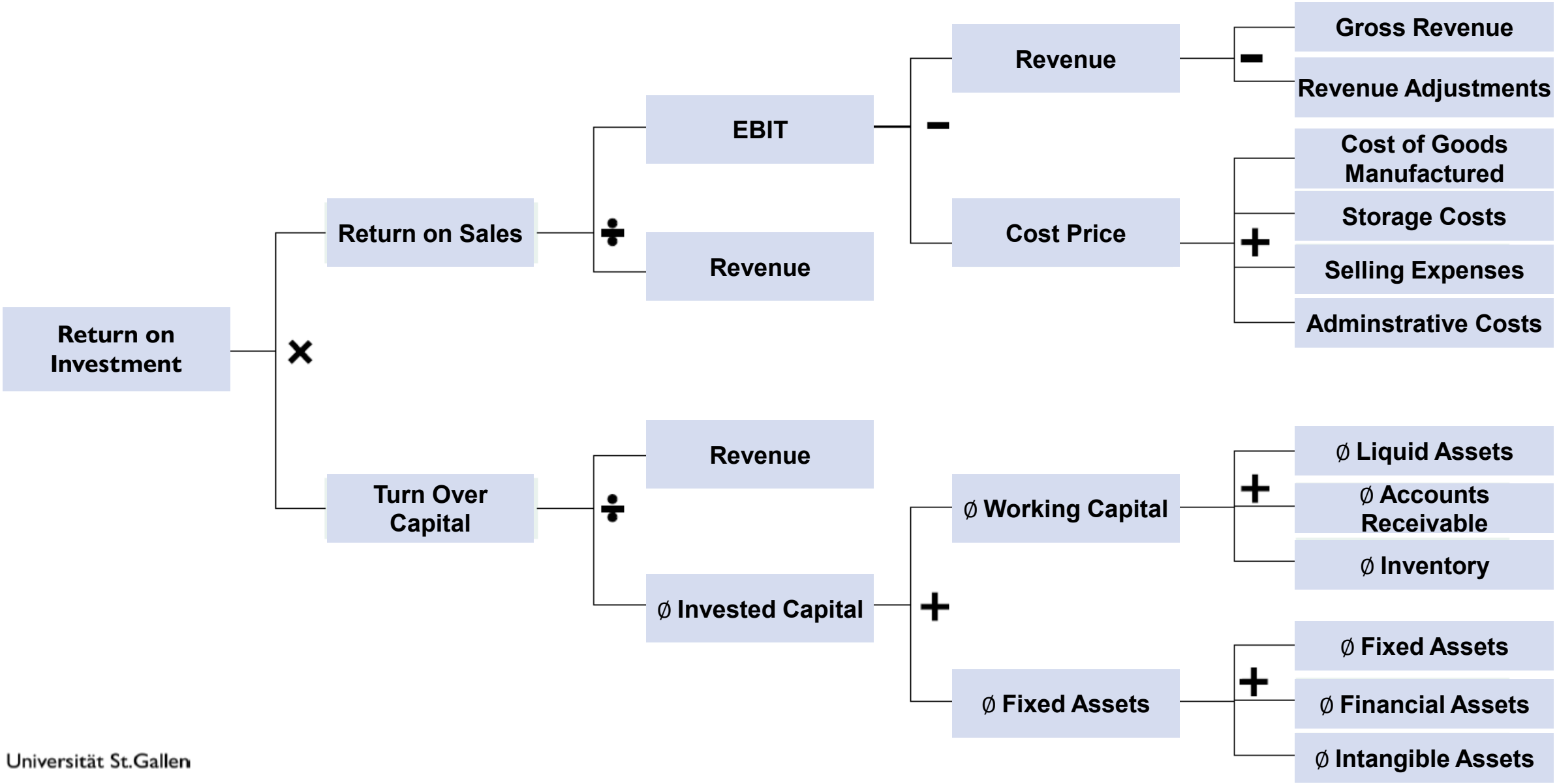
…and the DuPont-Scheme for Measuring Profitability



Task 4: ROI, RI, EVA

Hamilton Corp. is a reinsurance and financial services company. Hamilton strongly believes in evaluating the performance of its stand-alone divisions using **financial metrics such as ROI and residual income**. For the year ended December 31, 2024, Hamilton's CFO received the following information about the performance of the **property/casualty division**:

Sales revenues	CHF 900,000
EBIT	CHF 225,000
Total assets	CHF 1,500,000
Current liabilities	CHF 300,000
Debt (interest rate: 5%)	CHF 400,000
Common equity (book value)	CHF 500,000

For the purposes of divisional performance evaluation, Hamilton **defines investment as total assets** and **income as EBIT (that is, income before interest and taxes)**. The firm pays a **flat rate of 25% in taxes** on its income.

Task 4: ROI, RI, EVA

1. What was the **net income after taxes of the property/casualty division**?
2. What was the **division's ROI for the year**?
3. Based on Hamilton's **required rate of return of 8%**, what was the property/casualty division's residual income for 2024?
4. Hamilton's CFO has heard about **EVA and is curious about whether it might be a better measure to use for evaluating division managers**. Hamilton's four divisions have similar risk characteristics. Hamilton's **debt trades at book value** while its **equity has a market value approximately 150% that of its book value**. The company's **cost of equity capital is 10%.** Calculate each of the following **components of EVA for the property/casualty division**, as well as the final EVA figure:
    - Net operating profit after taxes
    - Weighted-average cost of capital
    - Investment, as measured for EVA calculations

Task 4: ROI, RI, EVA

1. What was the net income after taxes of the property/casualty division?

Sales revenues	CHF 900’000
EBIT	CHF 225,000
Total assets	CHF 1,500,000
Current liabilities	CHF 300,000
Debt (interest rate: 5%)	CHF 400,000
Common equity (book value)	CHF 500,000

For the purposes of divisional performance evaluation, Hamilton **defines investment as total assets** and **income as operating income (that is, income before interest and taxes)**.

The firm pays a **flat rate of 25% in taxes** on its income.

EBIT	**CHF 225’000**
Less:	
Interest expenses (CHF 400’000 * 5%)	**CHF 20’000**
Income after interest and before taxes	**CHF 205’000**
Income taxes (CHF 205’000 * 25%)	**CHF 51’250**
Net income after taxes	**CHF 153’750**

Return on Investment (ROI)

1

ROI is an accounting measure of income divided by an accounting measure of investment. ROI may be decomposed into its two components as follows:

$$\text{Return on Investment} = \frac{\text{EBIT}}{\emptyset\ \text{Invested Capital}} = \underbrace{\frac{\text{EBIT}}{\text{Revenue}}}_{\text{Return on Sales}} \times \underbrace{\frac{\text{Revenue}}{\emptyset\ \text{Invested Capital}}}_{\text{Turn Over Capital}}$$

This approach is known as the DuPont Method of Profitability Analysis. It recognizes the two basic ingredients in profit making: increasing income per dollar of revenue and using assets to generate more revenues

Task 4: ROI, RI, EVA

2. What was the division's ROI for the year?

Sales revenues	CHF 900'000
EBIT	CHF 225,000
Total assets	CHF 1,500,000
Current liabilities	CHF 300,000
Debt (interest rate: 5%)	CHF 400,000
Common equity (book value)	CHF 500,000

For the purposes of divisional performance evaluation, Hamilton **defines investment as total assets** and **income as operating income (that is, income before interest and taxes)**. The firm pays a **flat rate of 25% in taxes** on its income.

Direct Method

$$\text{ROI} = \frac{\text{CHF 225'000}}{\text{CHF 1'500'000}} = 15\%$$

Du-Pont Scheme

$$\text{ROS} = \frac{\text{CHF 225'000}}{\text{CHF 900'000}} = 25\%$$

x

$$\text{Turn Over Capital} = \frac{\text{CHF 900'000}}{\text{CHF 1'500'000}} = 0.6$$

15%

Residual Income (RI)

2

- Residual Income (RI) is an accounting measure of income minus a dollar amount for required return on an accounting measure of investment

Residual Income = Income − (RRR x Investment)

- Required rate of return (RRR) times the investment is the imputed cost of the investment
- The imputed cost of the investment is a cost recognized in particular situations but not recorded in financial accounting systems because it is an opportunity cost

Task 4: ROI, RI, EVA

3. Based on Hamilton's required rate of return of 8%, what was the property/casualty division's residual income for 2024?

Sales revenues	CHF 900'000
EBIT	CHF 225,000
Total assets	CHF 1,500,000
Current liabilities	CHF 300,000
Debt (interest rate: 5%)	CHF 400,000
Common equity (book value)	CHF 500,000

For the purposes of divisional performance evaluation, Hamilton **defines investment as total assets** and **income as operating income (that is, income before interest and taxes)**. The firm pays a **flat rate of 25% in taxes** on its income.

Imputed Cost of Investment

Investment		Required rate of return		
CHF 1'500'000	x	8%	=	CHF 120'000

Residual Income

Income	-	Imputed Income		
CHF 225'000		CHF 120'000	=	CHF 105'000

Economic Value Added (EVA)

EVA is a variation of RI used by many companies

EVA = After_tax Operating Income − [WACC x (Total Assets − Current Liabilites)]

Weighted average cost of capital equals the after-tax average cost of all long-term funds in use
EVA substitutes the following number in the RI calculation:

- Income: After-tax **operating** income
- Required rate of return: (After-tax) weighted-average cost of capital
- Investment: Total assets minus current liabilities

Task 3: ROI, RI, EVA

4. Calculate each of the following components of EVA for the property/casualty division, as well as the final EVA figure

Sales revenues	CHF 900'000
EBIT	CHF 225,000
Total assets	CHF 1,500,000
Current liabilities	CHF 300,000
Debt (interest rate: 5%)	CHF 400,000
Common equity (book value)	CHF 500,000

For the purposes of divisional performance evaluation, Hamilton **defines investment as total assets** and **income as operating income (that is, income before interest and taxes)**. The firm pays a **flat rate of 25% in taxes** on its income. **Take EVA-definition here instead, i.e. NOPAT.**

Net Operating Profit After Taxes (NOPAT)

EBIT		(1 – Tax rate)		
CHF 225'000	x	(1 – 0.25)	=	CHF 168'750

Task 3: ROI, RI, EVA

4. Calculate each of the following components of EVA for the property/casualty division, as well as the final EVA figure

Sales revenues	CHF 900'000
EBIT	CHF 225,000
Total assets	CHF 1,500,000
Current liabilities	CHF 300,000
Debt (interest rate: 5%)	CHF 400,000
Common equity (book value)	CHF 500,000

For the purposes of divisional performance evaluation, Hamilton **defines investment as total assets** and **income as operating income (that is, income before interest and taxes)**. The firm pays a **flat rate of 25% in taxes** on its income. **Take EVA-definition here instead, i.e. NOPAT.**

Market Value of Debt

CHF 400'000

After-Tax Cost of Debt

Interest rate		(1 – Tax rate)		
5%	x	(1 – 0.25)	=	3.75%

Market Value of Equity (Cost of Equity Capital = 10%)

Book value of equity		Market value = 150% of BV		
CHF 500'000	x	150%	=	CHF 750'000

Weighted-Average Cost of Capital (WACC)

$$\frac{(3.75\ \% * \text{CHF } 400'000) + (10\% * \text{CHF } 750'000)}{(\text{CHF } 400'000 + \text{CHF } 750'000)} = 7.83\%$$

Task 3: ROI, RI, EVA

4. Calculate each of the following components of EVA for the property/casualty division, as well as the final EVA figure

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For the purposes of divisional performance evaluation, Hamilton **defines investment as total assets** and **income as operating income (that is, income before interest and taxes)**. The firm pays a **flat rate of 25% in taxes** on its income. **<u>Take EVA-definition here instead, i.e. NOPAT.</u>**

Investment

Total Assets		Current liabilities		
CHF 1’500’000	-	CHF 300’000	=	CHF 1’200’000

EVA

NOPAT	(WACC * Investment)		
CHF 168’750	- (7.83 * CHF 1’200’000)	=	CHF 74’790